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EXTERNAL MALE INCONTINENCE CLAMP

Abstract

An incontinence clamp is provided. The incontinence clamp includes an upper clamp arm and a lower clamp arm. Each of the upper and lower clamp arms include a first end, a second end, an inner surface and an outer surface. The inner surfaces face each other. A hinge pivotally connects the first ends of the upper and lower clamp arms together. An upper guide is coupled to the inner surface of the upper clamp arm and a lower guide is coupled to the inner surface of the lower clamp arm. A connector releasably connects the second ends of the upper clamp arm and the lower clamp arm together.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This US continuation application claims the benefit of U.S. non-provisional application Ser. No. 18/596,942, filed Mar. 6, 2024, which is a continuation of U.S. non-provisional application Ser. No. 18/191,436, filed Mar. 28, 2023, issued as U.S. Pat. No. 11,931,235 on Mar. 19, 2024, which is a continuation of U.S. non-provisional application Ser. No. 16/810,148, filed Mar. 5, 2020, issued as U.S. Pat. No. 11,642,205, issued May 9, 2023, which is a continuation of Ser. No. 15/253,051, filed Aug. 31, 2016, issued as U.S. Pat. No. 10,247,728 on Apr. 21, 2020, the contents of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to incontinence prevention and, more particularly, to an external male incontinence clamp.

[0003] Incontinence is a problem encountered by many men and is more common among geriatrics. While the cause of male incontinence may have a variety of different reasons, the uncontrolled and undesired passage of urine is a significant problem resulting in embarrassment, restriction of activities and depression.

[0004] While adult diapers are one solution to the problem, they are bulky, retain the moisture, and are embarrassing and difficult for some aging men to handle. Other incontinence clamps have been developed over the years, but since anatomies is different between individuals, it is difficult to create a good fit between the device and the penis. This causes the devices to be extremely uncomfortable and ineffective.

[0005] Most of the incontinence clamps developed use foam pads as the interface material between the device and the skin. This creates irritation and swelling of the skin. When the foam collects urine, it is very difficult to clean and device smells bad. Some other clamps include Polivynilsiloxane pads. This material is much better than foam but given the properties of the material, it will release natural oil which acts as lubricant and causes the clamp to slip out of position.

[0006] As can be seen, there is a need for an improved male incontinence clamp which retains on the penis during everyday activities, that adapts to the anatomy of the patient, that is easy for older men to use and provides control of incontinence without causing irritation or swelling of the tissue.

SUMMARY OF THE INVENTION

[0007] In one aspect of the present invention, an incontinence clamp comprises: an upper clamp arm comprising a first end, a second end, an inner surface and an outer surface; a lower clamp comprising a first end, a second end, an inner surface and an outer surface; a hinge pivotably connecting the first end of the upper clamp arm to the first end of the lower clamp; an upper guide coupled to the inner surface of the upper clamp arm and comprising a curved concave inner surface; a lower guide coupled to the inner surface of the lower clamp and comprising a first side portion, a second side portion and a middle portion disposed in between the first side portion and the second side portion, wherein the middle portion comprises a curved convex portion protruding

towards the upper clamp arm; and a connector releasably connecting the second end of the upper clamp arm to the second end of the lower clamp arm.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following drawings, description, and claims. At least one specification heading is required.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of an embodiment of the present invention shown in a closed configuration;

[0010] FIG. 2 is a perspective view of an embodiment of the present invention shown in an open configuration;

[0011] FIG. 3 is an exploded view of an embodiment of the present invention;

[0012] FIG. 4 is a section view of the present invention taken along line 4-4 in FIG. 2; and

[0013] FIG. 5 is a section detail view of the present invention taken along line 5-5 in FIG. 1.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0014] The following detailed description is of the best currently contemplated modes of carrying out exemplary embodiments of the invention. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the invention, since the scope of the invention is best defined by the appended claims.

[0015] The present invention includes an external male incontinence clamp. When the clamp is placed on the penis, the device occludes the urethra while promoting blood circulation. The clamp of the present invention includes guides that adapt the user anatomy in shape and size. The material used is biocompatible, does not cause irritation or swelling, promotes a good perspiration of the skin and does not slip off of the penis.

[0016] Referring to FIGS. 1 through 5, the present invention includes an incontinence clamp. The incontinence clamp includes an upper clamp arm 10 and a lower clamp arm 18. Each of the upper and lower clamp arms 10, 18 include a first end, a second end, an inner surface and an outer surface. The inner surfaces face each other. A hinge 16 pivotally connects the first ends of the upper and lower clamp arms 10, 18 together. An upper guide 24 is coupled to the inner surface of the upper clamp arm 10 and a lower guide 26 is coupled to the inner surface of the lower clamp arm 18. A connector 21 releasably connects the second ends of the upper clamp arm 10 and the lower clamp arm 18 together.

[0017] The upper guide 24 and lower guide 26 contours to the penis 34 and applies pressure to the urethra 36, preventing incontinence. The upper guide 24 includes a curved concave inner surface. The lower guide 26 includes a first side portion, a second side portion and a middle portion disposed in between the first side portion and the second side portion. The middle portion includes a curved convex portion protruding towards the upper clamp arm 10. The first side portion and the second side portion may each include a curved concave inner surface curving in the opposite direction of the middle portion. The upper guide 24 and the lower guide 26 may be formed of a rigid polymer to provide appropriate pressure to the penis 34. For example, the rigid polymer is acrylonitrile butadiene styrene.

[0018] In certain embodiments, the upper guide 24 and the lower guide 26 are releasably attached to the inner surface of the upper clamp arm 10 and lower clamp arm 18 respectively. Since the upper and lower guides 24, 26 may be removable, a user may replace the guides 24, 26 depending on the size of the penis 34. In such embodiments, a slot 12 may be formed on the inner surface of the upper clamp arm 10. The upper clamp guide 24 includes a ridge releasably retained within the slot 12. A slot 20 may be formed on the inner surface of the lower clamp arm 18. The lower clamp

guide **26** includes a ridge releasably retained within the slot **20**.

[0019] The present invention may further include a padding **28, 30**. An upper padding **28** may be secured to the upper clamp arm **10** and a lower padding **30** may be secured to the lower clamp arm **18**. The padding **28, 30** may be in the shape of a cover and is sized to releasably secure over the upper and lower guides **24, 26** respectively. The padding **28, 30** may include ridges formed on an outer surface to prevent slipping. The padding **28, 30** adds comfort to the present invention. The padding **28, 30** may be made of a thermoplastic elastomer.

[0020] The connector **21** of the present invention may include a plurality of connecting positions. Each of the connecting positions forms a different internal diameter of the incontinence clamp. For example, one of the second ends includes an interlocking tooth **14** and the other of the second ends includes a plurality of interlocking teeth **22**. The interlocking tooth **14** is secured in between the plurality of interlocking teeth **22**, releasably connecting the second ends together. As illustrated in the Figures, the interlocking tooth **14** may protrude from the outer surface of the upper clamp arm **10**. The plurality of interlocking teeth **22** may protrude from an inner surface of a tooth tab **32** extending from the second end of the lower clamp arm **18**. Pressing the clamp arms **10, 18** together releasably retains the second end of the upper clamp arm **10** to the second end of the lower clamp arm **18**. The interlocking tooth **14** disposed in between different interlocking teeth **22** adjusts the diameter of the incontinence clamp.

[0021] In use, the users select the guides that better correspond to their penis shape and size. Then the user assembles the guides into the clamp by sliding the guides in the slots formed in the inner surface. After assembling the guides, the user assembles the pads over the guides. Once the pads are assembled, the user opens the incontinence clamp by releasing the catch and lifting up the top arm. The user may then place the penis between the pads with the incontinence clamp about halfway down the shaft. Latch the incontinence clamp to compress the urethra at the level that's comfortable to the user. The bottom part of the clamp is designed to concentrate the pressure in the urethra while providing room to the corpora cavernosa to move on the sides. The top part of the clamp is designed to promote a correct blood circulation and sensation, enabling the dorsal arteries, veins and nerve to be in a pressure significantly lower than the applied in the urethra. Given the variety of guides and pressure levels of the clamp, the adaptation between the device and the penis is much more effective, reliable and healthier for the patient. The pads with ridges permit the correct skin perspiration and prevent slippage. Due to the flexibility of the material, the pads provide comfort to the user.

[0022] It should be understood, of course, that the foregoing relates to exemplary embodiments of the invention and that modifications may be made without departing from the spirit and scope of the invention as set forth in the following claims.

Claims

1. A removable compression element for an incontinence clamp comprising: a guide dimensioned to span a majority of an inner surface of a movable clamp arm of the incontinence clamp, the movable clamp arm dimensioned to partially encircle a user's penis, the guide comprising a curved concave-inner surface, a convex outer surface, and a connector for coupling with movable clamp arm, and a padding attached to cover the guide, exclusive of the connector.
2. The removable compression element of claim 1, wherein the padding is releasably secured over the guide.
3. The removable compression element of claim 2, wherein the padding is formed of a thermoplastic elastomer.
4. The removable compression element of claim 2, wherein the padding comprises a plurality of ridges formed on an outer surface.
5. The removable compression element of claim 1, wherein the guide is formed of a rigid polymer.

6. The removable compression element of claim 5, wherein the rigid polymer is acrylonitrile butadiene styrene.

7. The removable compression element of claim 1, wherein the curved concave inner surface further comprises: a curved convex protrusion extending from a middle portion of the curved concave inner surface.

8. The removable compression element of claim 1, wherein the connector comprises a plurality of connecting positions, wherein each of the connecting positions couples with the movable clamp arm.

9. A compression element for an incontinence clamp comprising: a removable guide configured to releasably couple with an arcuate clamp arm of the incontinence clamp, the removable guide having a convex outer face that underlies a majority of a concave inner surface of the arcuate clamp arm, the removable guide comprising a concave inner face, and a padding that surrounds a majority of the removable guide between the convex outer face and the concave inner face.

10. The compression element of claim 9, wherein the removable guide is dimensioned to selectively adjust a size of the incontinence clamp to a user's penis.

11. The compression element of claim 10, wherein the padding is releasably secured over the removable guide.

12. The compression element of claim 11, wherein the padding is formed of a thermoplastic elastomer.

13. The compression element of claim 11, wherein the padding comprises a plurality of raised ridges protruding from an outer surface of the padding.

14. The compression element of claim 9, wherein the removable guide is formed of a rigid polymer.

15. The compression element of claim 14, wherein the rigid polymer is acrylonitrile butadiene styrene.

16. The compression element of claim 9 further comprising: a protrusion extending from a medial position of the concave inner surface of the removable guide, the protrusion dimensioned to exert a pressure against a urethra of a user when the incontinence clamp is in a closed position.
